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A link to your group's project workspace (a Git repository, Deepnote, or Google Colab), with your mentors given access

<https://github.com/FrianaEngineer/INFO-2950-Group-Project>

Considerations:

- Jobexposure.csv versus BLS OEWS (national, cross-industry) wage level and wage dispersion in respect to AI.
- Cornell class roster versus USAJOBS as comparison of federal job AI demand and top university curriculum
 - How has implementation of AI as a required skill reflected in the educational curriculum at top university

1. <https://www.bls.gov/oes/special-requests/oesm25nat.zip> (Friana) and O*NET(Fiona)

Description: Our first dataset idea examines whether occupations associated with AI related skills will differ in employment, wage level, and wage dispersion. We would combine national employment and wage data from the BLS with occupational skill information from O*NET. This would allow us to compare the mean, median, and percentile wages of occupations with different levels of AI related skill requirements.

Availability: The May 2025 National Occupational Employment and Wage Statistics dataset is available as a downloadable ZIP file from the U.S. Bureau of Labor Statistics and doesn't require an API key. This dataset contains approximately 1,400 rows identified by 6 digit Standard Occupational Classification (SOC) codes or OEWS-specific occupational codes. It provides national total employment, mean and median annual wages, and wage percentiles. The rows represent different occupational aggregation levels recorded in O_GROUP, including major, broad, and detailed occupations. We would filter the data to detailed occupations where available.

O*NET or the Occupational Information Network is a platform that hosts data that shares data about different occupations within the US labor market. This website is sponsored by the US Department of Labor which makes it much more credible due to the fact that they are government sponsored. They have

data covering 55,000 jobs across the U.S. economy. They compile and sort different task statements in order to decipher the different skills that are needed to perform that specific job. This data is regularly updated from a wide variety of sources and O*NET is partnered with RTI International and other companies that help with the supervision and verification of the data produced.

O*NET data availability: We would specifically use the [Software Skills file from the O*NET 31.0 Database](#). This file connects O*NET-SOC occupation codes with software-skill examples and identifies the technologies categorized as hot or in demand. It can be downloaded for free as a Excel or CSV without an API key. O*NET does also offers an API that requires registration and an API key but we would use the downloadable file instead of that.

Both O*NET and the BLS data sets use the SOC which will help us to compare the numbers of people that are in each occupation from the BLS data to the skills and attributes required from the O*NET data. This will help us to get a much better understanding of how many people are actually required to have knowledge of AI in order to perform their jobs. The BLS data set will also help us to get a better idea for the demographics of the people that are in AI related occupations.

Data quality and limitations: BLS is a trustworthy source because its a federal government agency. Its May 2025 wage and employment estimates are based on survey data collected from approximately 1.1 million establishments over 3 years. However, the data doesnt include self-employed workers and some occupational estimates may be unavailable. O*NET is also trustworthy because it is sponsored by the U.S. Department of Labor and collects information from sources such as workers, occupational experts, employer job postings and government programs. One limitation is that different O*NET occupations may have been updated at different times.

Plan for combining the datasets: We would use SOC codes to connect O*NET's occupational skill information with BLS employment and wage information. If O*NET divides a six-digit SOC occupation into more detailed occupations, we would group those detailed occupations under the corresponding six-digit SOC code when necessary. This would allow us to examine whether occupations associated with AI-related skills have different employment levels and wage distributions.

In order to identify the AI related occupations we would have to search the software skill names for a documented list of terms such as artificial intelligence, machine learning, natural language processing, generative AI and deep learning. We would then have to create some kind of indicator which would show whether each occupation is associated with at least one AI related software skill before comparing its employment and wage measures.

2. USAJOBS API <https://developer.usajobs.gov/api-reference/get-api-historicjoa>
Developer portal: <https://developer.usajobs.gov/>

This dataset consists of federal job announcements collected from the USAJOBS Historic JOA, API, maintained by the US Office of Personnel Management. Each observation is a job announcement posted to [USAJOBS.gov](https://www.usajobs.gov). We would assemble a table with one row per announcement containing the date, the occupational series, positional title, grade range, and advertised min and max salary, and hiring agency. From the descriptor we would derive an indicator on whether the posting mentions AI terms and whether

it is a requirement or preference. Unlike the BLS dataset this has genuine dates and allows us to examine change over time.

The data is retrieved through a free public API and they have stated that no authorization and authentication is required because the data is publicly available so there is no API key and no cost. We won't need to scrape the website for our data.

The website is .gov and the information is produced and provided by the federal government's own HR agency, and it's a mandatory posting venue for competitive-service federal vacancies. Its system of record, and postings are public-domain government records. It seems that the quality of the data is also well structured, and each position carries a numerical series. Dates are explicit. Some concerns are that this dataset only has federal jobs and AI requirements in government may differ systematically from the private sector. Another concern is that postings of jobs do not necessarily equate to hiring and advertised demand may not be an accurate representation of real demand.

3. Cornell Class Roster API classes.cornell.edu/content/SU17/api-details

Description: Datasets that describe how universities have changed their courses with the implementation of AI as a job skill requirement. An archive of courses over the years could help us obtain this information by analyzing how AI has impacted university curriculum and how AI has been incorporated into coursework and course objectives.

Data availability: Cornell Class Roster API classes.cornell.edu/content/SU17/api-details

This API provides access to Cornell's class roster from the fall of 2015 through fall of 2025. This dataset has the course number, course name, course description, if they satisfy any distribution requirements, course prerequisites, and last term offered.

The API is free and does not require an API key. All methods use GET requests and should follow the URL structure <https://classes.cornell.edu/api/version/method.responseFormat?parameters>. All parameters are also provided on the website linked in the data availability section.

Quality and Limitations: This data is being collected directly from Cornell university which is a trustable source as they are reporting on their own course catalog archives. There are no concerns about the quality of the data since it is scrapped from the Cornell course roster page by Cornell themselves. Some limitations of this dataset might be that it only provides information about the Cornell course catalog so it is not entirely representative of how all universities have updated their course catalog as a whole to account for the new AI skill requirements for hiring purposes.

Questions:

- If we were to compare this data to other datasets that include how job postings have started to implement AI skills as part of the required skills section, would that be holistic enough for a project considering its only a Cornell specific dataset?
- A job posting can cover multiple vacancies and agencies may vary in how they bundle. Should our unit be the announcement or the vacancy? If it is the announcement, it is easier to depend on the data but the result of our findings may not be as useful. If it is the vacancy, we might have to

make some assumptions based on the postings, but the results would be more useful and possibly joinable with other data.

- Cornell cross lists heavily, how would you advise to decide what unit to go by? (course, cross-listing, section). Course descriptions are often unchanged year by year despite content changes. Would you trust the course description as a proxy for content?
- For example if we were to join USAJOBS and Cornell Class rosters: since there is no shared key, would the join just be considered two parallel descriptive time series or would it be a defensible combination to compare the federal labor market to a university's curriculum?
- ONET uses ONET-SOC while OEWS uses SOC, so the join would be many to one. When we, if we do, collapse the ONET-SOC onto the SOC we would lose details on the occupation. And our average of the numeric information (such as salary) would be unweighted. Does this create a strong bias in our measurements and is there a better method of joining or aggregating that you would recommend?